

An Anatomy of Imported Inflation: Evidence from Colombian Firm-to-Firm Trade

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Preliminary. Do not circulate.

Abstract

Imported inflation depends not only on which products a country buys, but also on the trading relationships through which it buys them. Using the universe of Colombian customs declarations from 2001 to 2025, I construct a relationship-level import-price inflation index. The granular residual associated with unequal relationship weights has a covariance contribution equal to 63 percent of aggregate inflation variance. Most of this granularity arises because large relationships are concentrated in products with large common price swings. More than one-quarter, however, arises because their unit values move differently from those of other relationships importing the same product. During the 2014–16 peso depreciation, both margins on average dampened peso import-price inflation. Product-level indexes preserve granularity arising from exposure across products but, by construction, eliminate granularity within them.

Keywords: Imported inflation; granular fluctuations; firm-to-firm trade; import prices.

JEL codes: E31, F14, F41, L11.

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1 Introduction

Import prices are a key channel through which external shocks enter an open economy. Exchange-rate movements, commodity-price swings, foreign cost shocks, and supply disruptions first show up at the border, then feed into the prices paid by firms and households. This channel was especially important during the recent inflation surge, when higher import prices contributed materially to domestic inflation across countries (Bernanke and Blanchard, 2025; di Giovanni et al., 2024). Understanding where those price movements come from is therefore necessary both for measuring imported inflation and for assessing the appropriate policy response.

Most empirical work takes the product or country–product cell as the relevant unit of analysis. It decomposes import-price inflation into global, origin-specific, product-specific, and input–output components (Auer et al., 2019; Amiti et al., 2024; di Giovanni et al., 2024). But trade is carried out through relationships between foreign suppliers and domestic buyers. These relationships are often persistent, and import spending is highly unequal among them. Product-level aggregation may therefore conceal economically relevant variation across the relationships through which a given product is imported.

The importance of this variation depends on concentration. If expenditure is dispersed across many relationships, relationship-level price movements should wash out in the aggregate. If a small number of relationships carries large expenditure shares, they need not. This is the granular mechanism studied for output and trade and, more recently, for consumer prices (Gabaix, 2011; Di Giovanni et al., 2014; di Giovanni et al., 2022; Alvarez-Blaser et al., 2025). Related work shows that prices can differ across buyers within narrowly defined supplier–product cells (Fontaine et al., 2020; Alvarez et al., 2025). The question is whether such relationship-level variation is quantitatively important for aggregate import-price inflation, and through which margin.

I provide a bottom-up accounting of imported inflation at the supplier–buyer–product level. Granularity may arise through two margins. First, large relationships may be concentrated in products with large common price movements; I call this *granular product exposure*. Second, large relationships may experience price changes that differ from those of other relationships importing the same product; I call this *within-product relationship variation*. Product-level indexes preserve the aggregate effect of the first margin but obscure its relationship-level origin, while averaging out the second. I separate these two margins and quantify their contribution to aggregate import-price inflation.

I study these margins using the universe of Colombian customs declarations from 2001 to 2025. I define a relationship as a foreign supplier selling a specific HS10 product to a Colombian buyer, for a given supplier country and country of origin. The resulting monthly panel follows relationship-level unit values over nearly a quarter century, including the 2008–09 trade collapse, the 2014–16 peso depreciation, and the 2021–22 global inflation surge. The baseline sample consists of relationships observed in both the current quarter and the same quarter of the previous year, and inflation is measured as the four-quarter change in dollar unit values.

I first validate the customs-based price measures against Colombia’s official imported producer price index. In pesos, the index constructed from continuing relationships has a correlation of 0.89 with the official series. Aggregating the same transactions to the HS10 level yields a correlation of 0.88 and a nearly identical root-mean-squared error. Both customs indexes therefore reproduce the broad movements in independently collected import prices. The relationship-level index is useful not because it mechanically improves aggregate fit, but because it preserves the underlying heterogeneity needed for the decomposition below.

The main empirical fact is the concentration of import expenditure. The baseline sample contains about 79,000 continuing relationships in an average quarter, but the corresponding Herfindahl index is equivalent to only 541 equally sized relationships. The top 1 percent of relationships account for more than half of lagged import value. As a result, price changes at a relatively small number of relationships receive substantial weight in the aggregate index.

To quantify the importance of this concentration, I compare the value-weighted import-price index with an index that assigns equal weight to each continuing relationship. Their difference is the simple granular residual (Gabaix, 2011; Alvarez-Blaser et al., 2025). At the four-quarter horizon, the covariance of this residual with aggregate import-price inflation is equal to 63 percent of the variance of aggregate inflation. This covariance share measures the contribution of unequal relationship weights to aggregate price movements, showing that price changes at large relationships differ systematically from those at the average relationship and carry substantial weight in the aggregate index.

I then decompose aggregate inflation into three components: the price change of the average relationship, the product exposure of large relationships, and price variation across relationships within the same HS10 product. The decomposition applies the granular accounting of Di Giovanni et al. (2014), and its extension to import prices in

[Amiti et al. \(2024\)](#), at the relationship level. The corresponding covariance shares are 37, 46, and 17 percent. The last two components together constitute the 63 percent granular share. Granular product exposure is the larger margin: it accounts for nearly three-quarters of granular covariance because large relationships are disproportionately located in products with large common price swings. Within-product relationship variation contributes a further 17 percent, or more than one-quarter of granular covariance. Large relationships therefore not only sit in volatile products; they also reprice differently from other links importing the same narrowly defined product. Product aggregation retains the first source of granularity but removes the second.

The within-product estimate is robust to the construction of the product benchmark. It remains between 18 and 20 percent when I restrict the sample to products with multiple relationships or use leave-one-out product means. Its level changes with the definition of the comparison market. Using HS10–origin means lowers the share to 8 percent because movements common to suppliers from the same origin are absorbed by the benchmark. Using broader, time-consistent HS families raises it to 33 percent because price differences across narrower products are then assigned to the relationship residual. The invariant conclusion is that within-product variation survives aggregation, although its measured importance depends on where the product market is drawn.

The direction of these granular components is economically consequential. During the 2014–16 peso depreciation and global commodity-price decline, the simple granular residual and the within-product relationship component were negative in every quarter. Unequal relationship weights lowered dollar import-price growth by an average of 7.2 percentage points relative to the equal-weight index. Granular product exposure contributed -4.1 points and within-product price movements a further -3.1 points. Because the peso–dollar conversion is common across relationships, these gaps carry directly into peso import-price inflation. In this episode, large relationships therefore cushioned rather than amplified the border-price effects of depreciation. This contrasts evidence from the U.S. consumer-price index, where large firms have been shown to amplify aggregate inflation ([Alvarez-Blaser et al., 2025](#)).¹

Separating between commodities and differentiated goods reveals that the two product types have different granular patterns. The covariance share of the simple granular residual is 62 percent for commodities and 44 percent for differentiated goods. Commodity inflation

¹The difference is consistent with the fact that Colombia’s imports are concentrated in commodities, which experienced a large common price decline, whereas U.S. consumer prices are concentrated in differentiated goods, which experienced a large common price increase.

has a larger simple granular residual because large relationships are concentrated in volatile products. After common HS10 movements are removed, however, within-product relationship variation is larger for differentiated goods. Energy is an important exception, with substantial contributions from both product exposure and within-product variation. Commodities therefore matter mainly because of where large relationships are exposed, whereas differentiated-goods inflation depends more on how unit values move across relationships within the product. These are different sources of aggregate fluctuations and call for different empirical explanations.

Recent work establishes that large firms shape consumer-price inflation ([Alvarez-Blaser et al., 2025](#)). Building on this insight, a first contribution of this paper is to identify the bilateral trading relationship as a measurable unit of imported inflation and show that granularity at this level has two distinct sources. The exact accounting separates the effect of concentrating import expenditure in products with common price movements from the effect of unequal repricing within those products.

The paper also moves imported-inflation accounting below the country–product cell. Existing decompositions measure global, origin, product, and input-output components, but cannot determine whether a product’s aggregate price movement is shared across its underlying buyers and suppliers. Conversely, work on firm-to-firm trade shows that prices vary across buyers within narrowly defined supplier–product cells ([Fontaine et al., 2020](#); [Alviarez et al., 2025](#)), but does not ask whether those bilateral movements survive national aggregation. I connect the two literatures by showing which relationship-level movements receive enough expenditure weight to enter an aggregate border-price index. The contribution is therefore both a measurement result and an aggregation result: the organization of imports across bilateral links is part of how imported inflation is formed.

The monthly accounting does not assign the within-product component to a structural pricing mechanism. As a preliminary extension, I use yearly data for 2009–2020 and the two-sided bargaining model of [Alviarez et al. \(2025\)](#) to separate changes in model-implied markups from a residual marginal-cost component. For differentiated goods, model-implied markup changes account for about 2 percent of the covariance with relationship-specific annual price changes within product–origin–year cells. Most variation is assigned to the residual cost component. This evidence is consistent with the logic of [Burstein et al. \(2025\)](#): aggregate markups move when shocks reallocate market shares, whereas large price movements that leave bilateral shares largely unchanged need not generate large markup movements. I treat this exercise as a model-based interpretation of the

accounting results, separate from the monthly decomposition.

The rest of the paper is organized as follows. Section 2 describes the data and price indexes. Section 3 documents concentration, separates granular product exposure from within-product relationship variation, and compares commodities with differentiated goods. Section 4 presents the preliminary yearly markup decomposition.

2 Data and Measurement

This section describes the data, the construction of the monthly import panel, and the inflation measures. The unit of observation is a supplier–buyer–product relationship: a foreign exporter, a Colombian importer, and an HS10 product, holding fixed the supplier’s reported country and the product’s country of origin. Fixing both country fields prevents a change in a supplier’s recorded location or in the origin of the good from appearing as a price change within a continuing relationship. A relationship is *continuing* at horizon k if the complete cell is observed in both $t - k$ and t . I denote relationships by r throughout.

2.1 Data Sources and Panel Construction

The main data source is the universe of Colombian merchandise import declarations recorded by the customs authority, the *Dirección de Impuestos y Aduanas Nacionales* (DIAN). I use DIAN’s monthly statistical archives to construct and harmonize the import data from 2001 through 2025, at monthly, quarterly, and yearly frequencies.² Each declaration reports the Colombian importer, the foreign exporter’s name and reported country, the HS10 product, the country of origin, FOB value in dollars, and net weight in kilograms. The assembled matched panel contains 36.1 million relationship–month observations. In the median year, it contains approximately 755,000 active relationships, 84,600 foreign suppliers, 25,900 Colombian buyers, and 6,450 HS10 products. Appendix Table A.2 reports selected years.

Colombian importers are uniquely identified by their tax number (*Número de Identificación Tributaria*, NIT). Foreign exporters instead appear as free-text names, truncated at forty characters and subject to spelling variation and encoding inconsistencies. I construct longitudinal exporter identifiers using a matching procedure described in detail

²Aggregate FOB values in the assembled data match official DANE trade statistics year by year to within 0.01 percent whenever the published bulletins allow an exact comparison.

in Appendix D.1. The resulting identifiers cover nearly all import value in most years. Unassigned transactions enter aggregate totals but not the relationship-level analysis.

The sample spans six editions of the Harmonized System, from HS 1996 through HS 2022, and five transitions between them. To avoid spurious price changes or incorrect elimination of a valid price comparison at revision boundaries, I construct time-consistent product families in the spirit of [Pierce and Schott \(2009\)](#), whereby HS6 codes connected through official correspondence tables are assigned to a common family. The baseline retains raw HS10 codes as the market definition; the family measure provides a cross-revision alternative and necessarily defines a broader product market.

At the annual horizon, the matched sample contains about 240,000 continuing relationships on average. Continuing relationships carry a substantial share of import value despite high turnover, consistent with evidence from other countries ([Martin et al., 2023](#)); continuation coverage varies with the accounting horizon and is reported below.

Additional Data Sources Exchange rates are monthly averages of the official peso-dollar exchange rate, which I use to convert customs unit values from dollars into pesos. I also construct exchange rates between each origin country’s currency and the dollar to document exchange-rate movements across import origins.

I validate the customs-based measures against DANE’s monthly imported producer price index (IPPI). Because DANE collects transaction prices independently of the customs records, the IPPI provides an external benchmark that does not share their measurement error. I obtain Brent crude prices from FRED for the commodity-exposure analysis. I classify products using the conservative [Rauch \(1999\)](#) classification. Homogeneous and reference-priced goods are treated as commodities and divided into energy (HS 27), metals (HS 26 and 72–83), and agriculture (HS 01–24). All remaining classified products are treated as differentiated goods.

2.2 Variable Construction

For relationship $r = (i, j, h, c, o)$ and month m , the dollar unit value is $p_{rm}^{usd} = V_{rm}/Q_{rm}$, where V_{rm} is FOB value and Q_{rm} is reported net weight in kilograms. The country fields c and o denote the supplier’s reported country and the product’s country of origin. The corresponding peso unit value is $p_{rm}^{cop} = E_m^{cop/usd} p_{rm}^{usd}$, where $E_m^{cop/usd}$ is the monthly average exchange rate. Unit values can move because prices change, but also because

weight is misreported, the mix of goods changes within an HS10 code, or shipment timing changes. These concerns are most severe in thin cells. I limit them by aggregating shipments before differencing, measuring price changes over an annual horizon, and dropping absolute unit-value changes of three log points or more.³

Frequency and horizon. For relationship r in quarter q , the dollar unit value is

$$p_{rq}^{usd} = \frac{\sum_{m \in q} V_{rm}}{\sum_{m \in q} Q_{rm}}, \quad g_{rq} = \Delta_4 \log p_{rq}^{usd}. \quad (1)$$

The quarterly unit value aggregates the underlying monthly shipments before the four-quarter difference is taken. This procedure partly mitigates measurement error caused by quantity error or composition within an HS10 code.

The baseline calendar frequency is quarterly and the change horizon is four quarters. This preserves an annual inflation concept while using four observations per year, consistent with the subannual calendar-time and annual-change approach in [Alvarez-Blaser et al. \(2025\)](#) and [Amiti et al. \(2024\)](#). Quarterly aggregation also improves continuity relative to matching the same month in adjacent years. Continuing relationships retained after trimming account for 51 percent of current import value on average, but only 12 percent in the lowest-coverage quarter. The corresponding monthly twelve-month calculation covers 40 percent on average and 9 percent in the lowest month. This incomplete coverage is a substantive limitation: the results describe price changes among continuing, matched relationships and therefore exclude entry, exit, or unmatched trade.

Let $\mathcal{C}_q$ be the relationships observed in $q - 4$ and q , and let $\omega_{r,q-4}$ be relationship r 's share of lagged FOB value within that continuing sample. Baseline dollar import-price inflation is the geometric Laspeyres index

$$\pi_q = \sum_{r \in \mathcal{C}_q} \omega_{r,q-4} g_{rq}. \quad (2)$$

Predetermined weights prevent a current unit-value change from altering its own aggregation weight. The main accounting is in dollars, which removes the common peso-dollar conversion and isolates movements in border unit values. For validation against the domestic-currency IPPI, I convert relationship-level dollar unit values into pesos with the monthly peso-dollar exchange rate and form twelve-month indexes. For robustness, I

³Results are robust to thresholds of 2 and 2.5 log points.

repeat the accounting with monthly one-month changes, quarterly one-quarter changes, monthly twelve-month changes, and annual changes. The four-quarter series contains 96 overlapping quarterly observations.

3 Accounting for Imported Inflation

This section assesses what relationship-level unit values add to the measurement of imported inflation. I first compare the customs-based relationship index with an independently collected measure of import prices. I then examine whether the concentration of import value across relationships gives large suppliers and buyers a disproportionate role in aggregate inflation. Next, I distinguish price movements associated with the products traded by large relationships from movements that arise within narrowly defined products. Finally, I compare these margins across commodities and differentiated goods.

3.1 Bottom-Up Imported Inflation

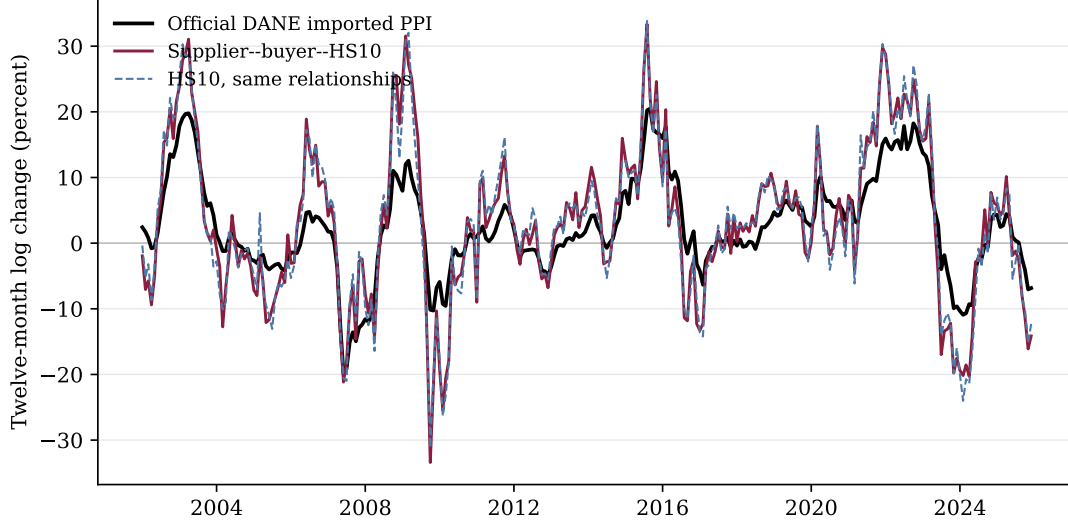
Customs unit values contain substantial information about import prices. I establish this by comparing customs-based indexes with DANE’s imported producer price index (IPPI), which is collected independently of the customs records.

Figure 1 compares twelve-month peso inflation in the IPPI with two Laspeyres indexes constructed from the same set of continuing, matched relationships. The relationship index computes a unit-value change for each link and aggregates those changes using lagged import-value weights. The product index first pools current and lagged values and weights across exactly those relationships within each HS10 product, and then aggregates product-level changes. Holding the underlying links fixed ensures that the comparison reflects the level at which their transactions are aggregated rather than differences in sample coverage.

The relationship index has a correlation of 0.89 with the official series over 288 months, compared with 0.88 for the same-sample HS10 index. Their root-mean-squared errors are 6.2 and 6.3 percentage points, respectively. Both customs indexes therefore recover the broad movements in official import-price inflation, while retaining relationships provides only a modest improvement in fit once the sample is held fixed. This exercise validates the customs-based measures but does not by itself establish the aggregate importance of within-product relationship variation. The accounting exercises below address that

question directly.

Figure 1: Customs indexes and official imported producer-price inflation



Notes. Twelve-month log changes in pesos. The relationship index forms unit values for continuing supplier–buyer–HS10 cells, holding fixed reported supplier country and origin, before aggregation. The product index first pools the values and weights of exactly those continuing relationships within HS10 products. Both customs indexes use lagged-value Laspeyres weights. The official series is DANE’s imported producer price index. The sample contains 288 monthly observations.

3.2 Concentration and Its Aggregate Relevance

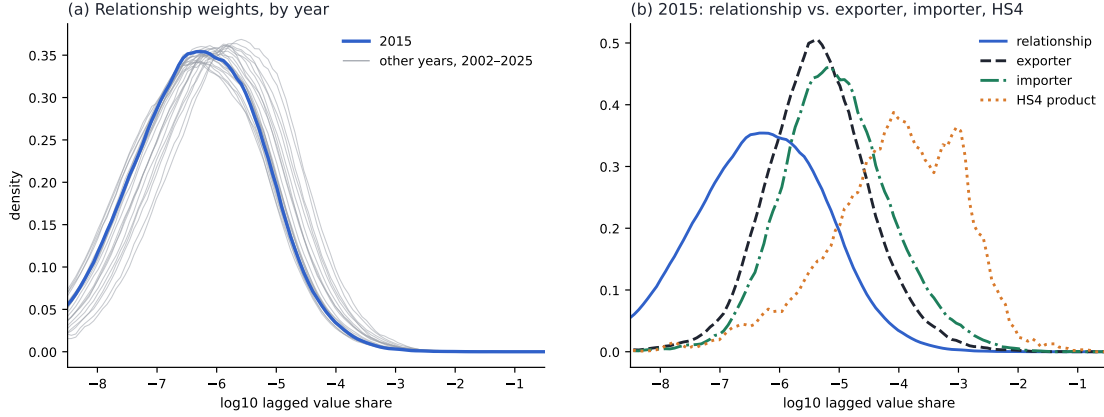
The aggregate importance of relationship-level price changes depends on how import expenditure is distributed across relationships. If expenditure were spread evenly across many links, unusually large price changes at individual relationships would receive little weight and wash out in the aggregate.

Colombian imports are concentrated in a small upper tail of large relationships. Figure 2 shows the distribution of lagged import-value shares. Panel (a) plots the distribution across continuing relationships by year. Relationship size spans more than six orders of magnitude and is strongly right-skewed. The shape of the distribution is also remarkably stable: the distribution in 2002 closely resembles that in 2025. Concentration is therefore a persistent feature of the import network, even though the identities of the underlying relationships change over time.

Panel (b) compares relationships with exporters, importers, and HS4 products in 2015. Most relationships are small relative to firms and product categories, but their

upper tails are similar. The largest relationships attain import shares comparable to those of the largest exporters and importers and, in the far tail, to entire HS4 product categories. In the 2015 matched sample, the top 1 percent of relationships account for 59 percent of lagged import value, compared with 56 percent for exporters and 52 percent for importers. Import expenditure is therefore concentrated at the level of individual supplier–buyer–product links, over and above its concentration across firms and products.

Figure 2: Distribution of Import-Value Weights



Notes. Densities of log10 lagged value shares in the quarterly matched samples, pooled within year. Panel (a) plots relationship shares, one line per year over 2002–2025 with 2015 highlighted. Panel (b) aggregates the same matched lagged values to the indicated unit within each quarter of 2015. Relationships are supplier–buyer–HS10 cells that also hold fixed reported supplier country and origin.

Two useful metrics to summarize this concentration are the Herfindahl index and its inverse, the effective number of relationships:

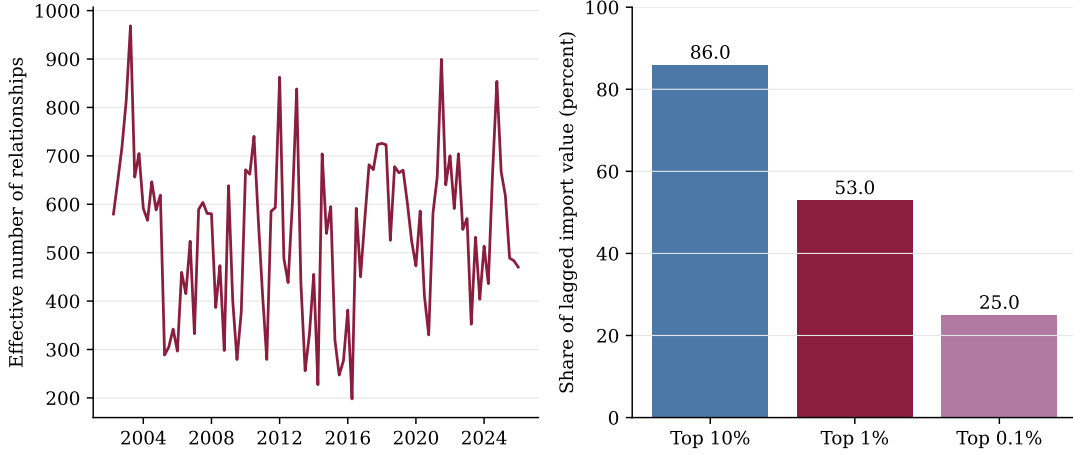
$$H_q = \sum_{r \in \mathcal{C}_q} \omega_{r,q-4}^2 \quad \text{and} \quad N_q^{eff} = H_q^{-1}, \quad (3)$$

where N_q^{eff} is the number of equally sized relationships that would generate the observed Herfindahl index. The baseline quarterly sample contains about 79,000 continuing relationships with usable four-quarter price changes. Their effective number averages only 541, with a median of 575.

Figure 3 shows the evolution of this concentration over time, along with the cumulative lagged-value shares held by the top 1, 5, and 10 percent of relationships. The concentration measure persists over time. The effective number ranges from about 200 to 970 without a trend and remains small relative to the number of continuing relationships. The import

shares held by the upper tail are similarly stable. Concentration is thus a structural feature of Colombian imports.

Figure 3: Concentration among Continuing Import Relationships



Notes. The left panel reports the inverse Herfindahl index of lagged relationship value in each quarterly matched sample. The right panel reports time averages of the cumulative lagged-value shares held by the indicated upper tails. Relationships are supplier–buyer–HS10 cells, also holding fixed supplier country and origin. Price changes are measured over four quarters.

Concentration gives large relationships the potential to move aggregate import-price inflation. Their actual contribution depends on whether their price changes differ from those of smaller relationships. I measure this margin using the simple granular residual (Gabaix, 2011; Alvarez-Blaser et al., 2025):

$$\Gamma_q^s = \underbrace{\sum_{r \in \mathcal{C}_q} \omega_{r,q-4} g_{rq}}_{\text{inflation for a dollar of imports}} - \underbrace{\frac{1}{N_q} \sum_{r \in \mathcal{C}_q} g_{rq}}_{\pi_q^{uw}: \text{inflation at the average relationship}}. \quad (4)$$

The first term is the value-weighted import-price index. The second is the average price change across continuing relationships, assigning each relationship equal weight. Their difference isolates the aggregate effect of the unequal size of relationships. It rises when large relationships experience greater price increases than the average relationship and falls when they experience smaller increases or larger declines.

At the four-quarter horizon, the correlation between Γ_q^s and aggregate dollar import-price inflation is 0.92. Its covariance with aggregate inflation equals 63 percent of the variance of aggregate inflation. The unequal weighting of relationships therefore accounts for a large share of aggregate price fluctuations. Large relationships matter not only

because they receive large expenditure weights, but because their price changes differ systematically from those of the average relationship.

This covariance can arise through two margins. Large relationships may import products subject to large common price movements, or they may experience different price changes from other relationships importing the same product. The next subsection separates these channels.

3.3 Product exposure and relationship-specific price movements

To separate the two channels, I write each relationship's price change as

$$g_{rq} = \delta_{hq} + u_{rq}, \quad \text{with} \quad \delta_{hq} = \frac{1}{N_{hq}} \sum_{r \in (h,q)} g_{rq}, \quad (5)$$

where h denotes the HS10 product and δ_{hq} is the unweighted mean price change among relationships importing product h in quarter q . This mean captures movements common to the product without allowing its largest relationships to determine the benchmark. The residual u_{rq} measures how much relationship r 's price change differs from that benchmark. This additive decomposition follows the granular accounting in [Di Giovanni et al. \(2014\)](#) and its application to import prices in [Amiti et al. \(2024\)](#).

Aggregating both terms with lagged import-value weights yields the exact identity

$$\pi_q = \underbrace{\sum_r \omega_{r,q-4} \delta_{h(r)q}}_{P_q: \text{ common product}} + \underbrace{\sum_r \omega_{r,q-4} u_{rq}}_{R_q: \text{ relationship-specific}}. \quad (6)$$

The common product component, P_q , captures the value-weighted movements of the products in which import value is concentrated. The relationship component, R_q , captures the value-weighted deviations from the price change of the typical relationship importing the same HS10 product. I measure each component's contribution as its covariance with π_q divided by $\text{Var}(\pi_q)$. Because $P_q + R_q = \pi_q$, the two covariance shares sum to one.

Equation (6) also yields the exact decomposition of the simple granular residual. Because the product mean is unweighted across relationships, the unweighted average relationship change π_q^{uw} is also the unweighted average of the product means assigned to

those relationships. Define granular product exposure as $G_q^P = P_q - \pi_q^{uw}$. Then

$$\pi_q = \underbrace{\pi_q^{uw}}_{\text{average relationship}} + \underbrace{G_q^P}_{\text{granular product exposure}} + \underbrace{R_q}_{\text{within-product relationship}}, \quad \Gamma_q^s = G_q^P + R_q. \quad (7)$$

The first term is the price change of the typical relationship. The second captures whether large relationships are disproportionately exposed to products whose common price movements differ from those of the typical relationship. The third captures whether large relationships move differently from other relationships importing the same product. The last two terms are the two channels through which unequal relationship size enters aggregate inflation.

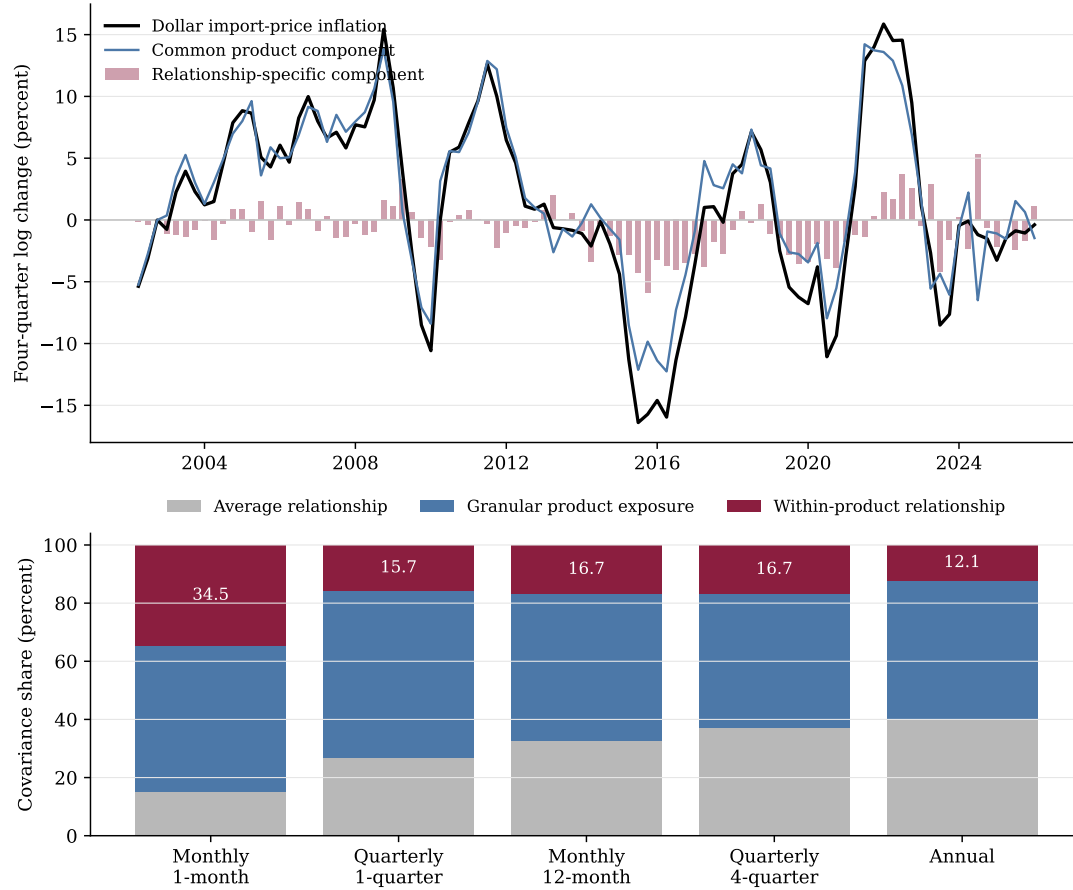
The relationship-specific component is a residual within the HS10 product cell. It may capture bilateral pricing, contract or delivery terms, differences in quality or shipment composition, timing, or measurement error. The decomposition therefore speaks only to how much price variation remains within narrowly defined products and whether that variation survives aggregation.

The upper panel of Figure 4 plots aggregate import-price inflation together with the two components. Common HS10 price movements account for most of the broad variation in the aggregate series. The relationship-specific component is smaller, but it is not negligible. It moves the aggregate away from the common product component in both inflationary and deflationary periods and is present throughout the sample rather than being concentrated in a particular episode.

Over the 96-quarter baseline, common HS10 movements account for 83.3 percent of the covariance with aggregate inflation, while within-product relationship movements account for 16.7 percent. Both shares are expressed relative to the variance of total inflation; equation (7) separates out the granular part. The three terms in equation (7) have covariance shares of 37.1 percent for the average relationship, 46.1 percent for granular product exposure, and 16.7 percent for within-product relationship variation. The last two sum to the 62.9 percent simple granular covariance share. Thus, 73 percent of granular covariance arises because large relationships are concentrated in products with unusually large common price movements; a further 27 percent arises within HS10 products. Product-level aggregation retains the first channel but eliminates the second.

The covariance shares summarize importance over the full sample but not the sign of each component in particular episodes. During the 2014–16 peso depreciation and

Figure 4: Product and relationship components of dollar import-price inflation



Notes. The upper panel reports four-quarter log changes observed quarterly. The common product component and relationship-specific component sum to the value-weighted index. The lower panel reports the three covariance shares in equation (7) under alternative calendar frequencies and change horizons. Granular product exposure and the within-product relationship component sum to the simple granular covariance share. All specifications use lagged import-value weights and continuing cells.

global commodity-price decline, the simple granular residual and the within-product relationship component were negative in every quarter; granular product exposure was negative in eleven of the twelve quarters. The value-weighted dollar index grew by an average of 7.2 percentage points less than the equal-weight index. Granular product exposure contributed -4.1 points and within-product relationship variation a further -3.1 points. Converting the two indexes into pesos leaves these differences unchanged because the same peso-dollar movement enters every relationship. Large relationships therefore cushioned peso import-price inflation during this episode, both through the products to which they were exposed and through their price movements relative to

other relationships within those products. This result is descriptive: it does not identify why large relationships adjusted differently, and the granular components need not be negative in other episodes.

Two choices affect the size of the relationship component: how the product mean is constructed and how narrowly the product market is defined. Table 1 examines both.

I first address the construction of the product mean. With a single relationship in a product cell, the product mean coincides with that relationship’s price change. More generally, in small cells, a relationship receives substantial weight in its own benchmark. These features are not important quantitatively. Restricting the sample to products with at least two or at least five continuing relationships yields relationship covariance shares of 17.7 and 19.8 percent, respectively, where both the component and aggregate index are reconstructed within the retained sample. Using leave-one-out product means yields a share of 19.7 percent. The baseline estimate is therefore little affected by singleton products or by the inclusion of a relationship in the mean against which it is compared.

I next vary the definition of the product market. In the baseline, the common component is defined at the HS10 level. As a result, price movements shared by suppliers of the same product and origin remain in the relationship component. Defining the common component instead at the HS10–origin level lowers the relationship share from 16.7 to 8.2 percent. Roughly half of the baseline relationship component is thus common to suppliers within the same product–origin cell, while the other half reflects variation within such cells.

The time-consistent HS families provide a broader product definition. They link HS6 codes across revisions of the classification, which is useful for comparisons over time, but they combine products that are distinct under the baseline HS10 definition. Price differences across these narrower products are therefore assigned to the residual. The relationship share rises to 33.1 percent under the family definition and to 24.8 percent when origin is also included in the common component. These estimates reflect the broader benchmark rather than stronger evidence of bilateral pricing.

The main conclusion is unchanged. Relationship-level price variation remains after conditioning on narrowly defined product movements, although its measured contribution depends on the level at which the common price component is defined.

Table 1: Relationship-specific covariance shares: definitions and robustness

Definition	Covariance share	Correlation	Periods
HS10 baseline	16.7	0.63	96
HS10, cells with at least two relationships	17.7	0.64	96
HS10, cells with at least five relationships	19.8	0.65	96
HS10, leave-one-out product mean	19.7	0.62	96
HS10 $\times$ origin common component	8.2	0.44	96
HS-family robustness	33.1	0.76	96
HS-family $\times$ origin common component	24.8	0.66	96

Notes. Four-quarter dollar unit-value changes observed quarterly. Covariance shares are $\text{Cov}(R_q, \pi_q) / \text{Var}(\pi_q)$. The correlation column reports $\text{Corr}(R_q, \pi_q)$. Cell-size restrictions reconstruct both R_q and π_q with weights renormalized within the retained sample. The leave-one-out row removes each relationship from its product mean and uses the corresponding sample of products with multiple relationships. HS families bridge product codes across Harmonized System revisions.

A final robustness check concerns calendar frequency and the horizon over which price changes are measured. Relationship-level movements may be short-lived and disappear with temporal aggregation. At the same time, higher-frequency estimates may be more sensitive to shipment timing and sparse cells. The lower panel of Figure 4 and Appendix Table A.1 report the decomposition across frequencies and horizons.

The relationship share is 34.5 percent for one-month changes measured monthly and 15.7 percent for one-quarter changes measured quarterly. At the annual horizon, however, the estimates are essentially identical across the two subannual frequencies: 16.7 percent for twelve-month changes measured monthly and 16.7 percent for four-quarter changes measured quarterly. The share falls to 12.1 percent when transactions are first aggregated to the annual level. Temporal aggregation therefore removes some relationship-level variation, but the annual-horizon result is not driven by the use of monthly rather than quarterly data.

These results also motivate the quarterly, four-quarter baseline. It measures inflation over an annual horizon, preserves more continuing relationships than the monthly specification, and does not select the frequency with the largest relationship component. Because four-quarter price changes overlap, the covariance decompositions are descriptive; statistical inference would need to allow for the resulting serial correlation.

3.4 Commodities: concentration versus product exposure

Commodities provide a useful benchmark for the relationship-level decomposition. If import prices are largely set in centralized world markets, relationships importing the same narrowly defined commodity should exhibit similar price changes. Yet commodity imports are also concentrated in large and volatile product categories. A large granular residual may therefore arise from the product exposure of large relationships even when little price variation remains within products.

Table 2 repeats the granular decomposition within the product groups defined in Section 2, with weights renormalized within each group. Commodities account for 32.8 percent of lagged import value in the matched sample and differentiated goods for 58.9 percent; the remaining 8.3 percent is unclassified. For commodities, the 62.2 percent simple granular covariance share consists of 47.9 percentage points of granular product exposure and 14.3 points of within-product relationship variation. For differentiated goods, the corresponding 44.3 percent share consists of 18.4 points of product exposure and 25.9 points of within-product variation. The two forms of granularity therefore differ sharply across the groups.

The larger simple granular residual for commodities mainly reflects the products in which large relationships are concentrated. Conditional on the product, unit-value movements differ more across relationships for differentiated goods. This is consistent with a larger role for bilateral terms, varieties, or contract-specific adjustment, although the decomposition does not distinguish among these channels.

There is substantial heterogeneity within commodities. The relationship share is 24.5 percent for energy, 7.8 percent for metals, and 1.5 percent for agricultural commodities. Agricultural prices are therefore close to the case in which a common product movement dominates. Energy is different: nearly one-quarter of its covariance with group inflation remains within HS10 product cells. This variation may reflect contract terms and bilateral pricing, but also differences in grade, cargo composition, transport arrangements, or quantity measurement within an HS10 code. The result should therefore be read as evidence of within-product variation in measured energy prices, rather than as evidence for a particular pricing mechanism.

Table 2: Import-price accounting by product group

Group	Value share	Product exposure	Within product	Simple granular	Corr.
Commodity	32.8	47.9	14.3	62.2	0.54
Differentiated	58.9	18.4	25.9	44.3	0.65
Energy	5.7	52.6	24.5	77.1	0.49
Metals	3.7	33.9	7.8	41.7	0.51
Agriculture	8.5	54.3	1.5	55.8	0.10

Notes. Four-quarter dollar unit-value changes observed quarterly. Products are classified using the conservative [Rauch \(1999\)](#) classification. Value shares are averages in the matched accounting sample; the omitted share is unclassified. Product exposure and within-product relationship variation are the two terms in the within-group simple granular residual and therefore sum to the simple granular covariance share. Energy, metals, and agriculture are subsets of commodities and do not exhaust that category. “Corr.” is the correlation between the within-product component and group inflation.

The distinction between the simple granular residual and the within-product relationship component is important. For commodities, the large granular residual reflects mainly the concentration of import value in relationships exposed to products with large common price movements. Once these product-level movements are removed, the relationship component is larger for differentiated goods, with energy as the main exception. The two margins are therefore distinct. Large commodity relationships matter for aggregate imported inflation because of the products they import, while price changes also differ across supplier–buyer relationships within narrowly defined products. The latter is not primarily a commodity phenomenon.

Taken together, the results clarify what is gained by retaining the relationship dimension. Customs unit values at both the relationship and product levels recover the broad movements in the official import-price index. Import expenditure is sufficiently concentrated that unit-value changes at large relationships affect the aggregate index. Nearly three-quarters of the granular covariance comes from the exposure of large relationships to products with large common price movements. A further one-quarter arises within HS10 products and is robust to alternative treatments of thin cells, although its magnitude depends on the definition of the product market. The signs of both components vary over time: during the 2014–16 depreciation, they cushioned peso import-price inflation. Product-level aggregation preserves granular exposure across products but eliminates within-product granularity by construction.

4 Interpreting Relationship-Specific Repricing

The monthly accounting isolates price movements at the relationship level without taking a stand on their source. A relationship’s price can depart from the product-level average because its supplier faces a different marginal-cost shock, because the bilateral markup changes, or because of compositional or quality differences within the HS10 code. Setting the latter aside, the two margins—marginal cost and markup—have different implications for the interpretation of granularity. Marginal-cost shocks are common to all relationships involving the same exporter supplying the same product, but they may be transmitted into the aggregate index in a concentrated market even if markups are unchanged. Markup shocks instead require a change in the division of surplus within the relationship.

The distinction also bears on the recent debate over profits and inflation. Markup increases by large firms have been proposed as an amplification mechanism in the post-pandemic episode, the so-called “sellers’ inflation” or greedflation ([Weber and Wasner, 2023](#); [Konczal and Lusiani, 2022](#)), while accounting and industry studies find a limited role ([Glover et al., 2023](#); [Conlon et al., 2026](#)). That literature works with indirect measures: profit shares in national accounts, industry markups estimated from financial statements, or calibrated pass-through. The customs data support a more direct measurement. Markups are computed at the level at which they are negotiated, the individual supplier–buyer relationship, from observed bilateral shares, and their contribution to border-price inflation is evaluated relationship by relationship. This is the paper’s distinct contribution to that debate: an inflation decomposition in which the markup is a measured object at the transaction level rather than an aggregate residual.

For this analysis, I use a yearly panel covering 2009–2020.⁴ Within the yearly panel, I remove movements common to an HS10 product, origin country, and year. The remaining variation compares relationships importing the same product from the same origin in the same year. It is therefore a narrower version of the relationship-specific component in the monthly analysis, which removes only the HS10–quarter mean.

The decomposition uses the two-sided bargaining model of [Alviarez et al. \(2025\)](#). A relationship r connects a foreign supplier i and a Colombian buyer j in an HS10 product h . The price is a markup over the supplier’s marginal cost, $p_{rt} = \mu_{rt} mc_{rt}$, with marginal

⁴The exercise has not yet been implemented on the 2001–2025 monthly panel and is not part of the baseline accounting in Section 3.

cost common across the buyers a supplier serves within a product. This restriction is what identifies markups from prices. The bargained markup has a closed form in two observed shares,

$$\mu_{rt} = \mu(s_{rt}, x_{rt}; \varphi, \theta), \quad (8)$$

where s_{rt} is supplier i 's share in buyer j 's expenditure on product h , and x_{rt} is buyer j 's share in supplier i 's units sold of the same product. The markup is increasing in s and decreasing in x : a supplier that accounts for a large share of its buyer's purchases charges a higher markup, and a buyer that absorbs a large share of its supplier's sales negotiates a lower one.

The parameters are the bargaining weight φ and the inverse export supply elasticity θ ; in this preliminary implementation I take both from [Alviarez et al. \(2025\)](#). Given the parameters, observed shares map directly into bilateral markups, and changes in shares into markup changes. For each continuing annual relationship, I then use the identity

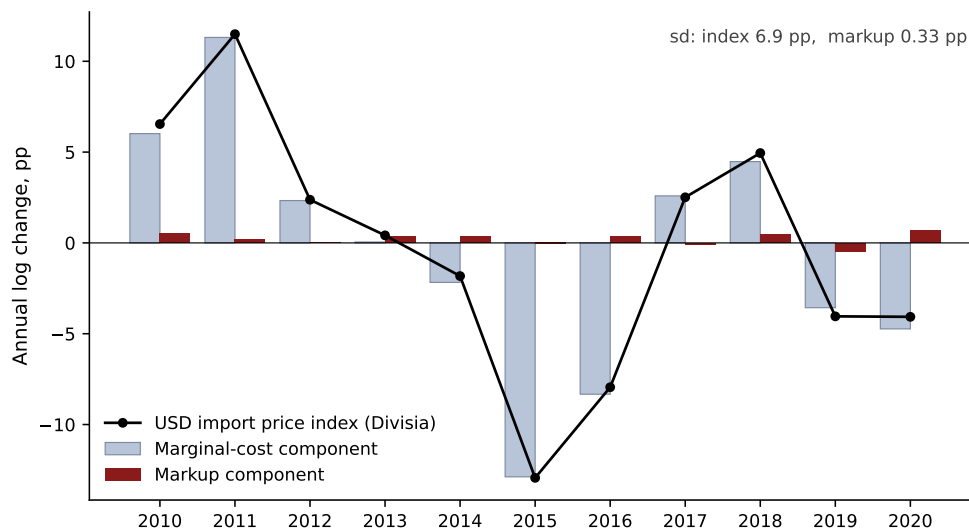
$$\Delta \log p_{rt} = \Delta \log \mu_{rt} + \Delta \log mc_{rt} \quad (9)$$

to obtain marginal-cost growth as the residual between the observed price change and the model-implied markup change. I residualize all three terms with respect to HS10–origin–year effects and aggregate their covariance with price changes using lagged import-value weights. By construction, the markup and marginal-cost covariance shares sum to one.

In differentiated goods, model-implied markup changes account for 1.6 percent of the covariance with annual relationship-specific price changes. The marginal-cost residual accounts for the remaining 98.4 percent. The result is economically stark: almost all annual repricing among relationships importing the same differentiated product from the same origin is assigned to the cost side of the price identity.

Figure 5 reports the aggregate counterpart. It plots the Divisia decomposition of annual changes in the dollar import price index into the model-implied markup component and the marginal-cost component. The index moves with a standard deviation of 6.9 percentage points per year; the markup component has a standard deviation of 0.3 percentage points and a covariance share of about one percent. In 2015, the largest depreciation year in the sample, the dollar index fell by 12.9 percentage points, of which the markup component contributed -0.06 . Years of large import-price movements are years of marginal-cost and exchange-rate movements; measured bilateral markups are essentially flat throughout.

Figure 5: Markup and Marginal-Cost Components of Import Inflation, Yearly Panel



Notes. Divisia decomposition of annual changes in the dollar import price index, yearly 2009–2020 panel. The markup component aggregates model-implied changes in bilateral markups with lagged value weights; the marginal-cost component is the residual in equation (9) and includes source-country exchange-rate movements. Bilateral markups are computed from observed bilateral shares using the bargaining model of [Alviarez et al. \(2025\)](#).

These results should be interpreted as model-based evidence rather than as direct measures of marginal costs. Any price movement not accounted for by the model-implied markup enters the residual, including freight, changes in quality or composition within HS10 products, unit-value measurement error, and misspecification of the bargaining model. The corresponding estimates for commodities are sensitive to product classification and individual years and are therefore not reported. The conclusion is narrower: in the annual data for differentiated goods, changes in observed bilateral shares imply too little markup variation to account for much of relationship-specific repricing.

A reduced-form exercise provides a complementary test. If the differential repricing of large relationships reflected changes in bilateral bargaining conditions, it should vary with observables related to those conditions, such as relationship age, exclusivity, and the bilateral shares that determine model-implied markups. Table 3 regresses annual changes in dollar unit values on the interaction between an indicator for large relationships—the top 0.5 percent by lagged import value—and an indicator for years in which the peso depreciates by more than five log points. I then add measures of relationship age, exclusivity, and bilateral shares, while progressively tightening the fixed effects from HS10–year to HS10–origin–year and, for supplier–products serving multiple Colombian

buyers, exporter–HS10–year.

In depreciation years, large relationships reduce dollar prices by an additional 3.3 percentage points relative to other relationships within the same HS10–year cell, and by 2.4 percentage points within HS10–origin–year cells. The differential disappears with exporter–HS10–year fixed effects, which compare buyers purchasing the same product from the same supplier in the same year. By contrast, interactions with relationship age, exclusivity, and bilateral bargaining shares are small and imprecisely estimated, and including them leaves the size effect largely unchanged. The pattern therefore points to variation common to a supplier–product rather than to changes in the bilateral characteristics of the match.

The stability of markups reflects the stability of market structure in firm-to-firm trade. In the model, the bilateral markup depends only on the shares s_{rt} and x_{rt} in equation (8), both of which are highly persistent. Within HS10 products, the value-weighted year-over-year correlation of the Herfindahl index is 0.83 for supplier shares and 0.92 for buyer shares, while the median absolute annual change in either index is below 0.05. The product-level markup implied by these shares varies within a range of about 3 log points over 2009–2020, and its annual changes have a standard deviation of 1.3 percentage points, compared with 6.9 percentage points for the dollar import-price index. Large movements in border prices therefore occur with relatively little change in the bilateral shares that determine markups. Shocks to exchange rates, world prices, or supplier costs can move transaction prices while leaving relative purchase and sales shares largely unchanged.

This is consistent with the logic of bottom-up models of markup fluctuations ([Burstein et al., 2025](#)), in which aggregate markups move when shocks reallocate market shares. Idiosyncratic shocks can generate such reallocations; shocks that are common within a product need not. In the Colombian data, the shocks associated with large movements in import prices appear to induce little reallocation across relationships and, correspondingly, little movement in model-implied markups. The comparison therefore suggests that the markup response depends more on how shocks affect relative market shares than on the size of the underlying price movement.

Table 3: What does relationship size capture?

Specification	Fixed effects	Coefficient	Estimate
A size only	HS10 x year	Top relationship	-0.002 (0.004)
A size only	HS10 x year	Top relationship x depreciation	-0.033*** (0.005)
A size only	HS10 x origin x year	Top relationship	-0.002 (0.003)
A size only	HS10 x origin x year	Top relationship x depreciation	-0.024*** (0.007)
A size only	exporter x HS10 x year	Top relationship	-0.012** (0.005)
A size only	exporter x HS10 x year	Top relationship x depreciation	-0.000 (0.009)
B size vs longevity	HS10 x origin x year	Top relationship x depreciation	-0.027*** (0.007)
B size vs longevity	HS10 x origin x year	Long relationship x depreciation	0.009* (0.005)
B size vs longevity	exporter x HS10 x year	Top relationship x depreciation	-0.002 (0.009)
B size vs longevity	exporter x HS10 x year	Long relationship x depreciation	0.006 (0.006)
C size vs exclusivity	HS10 x origin x year	Top relationship x depreciation	-0.024*** (0.007)
C size vs exclusivity	HS10 x origin x year	Exclusive supplier-product x depreciation	0.007 (0.004)
D bargaining shares	HS10 x origin x year	Buyer share of supplier-product x depreciation	0.009 (0.005)
D bargaining shares	HS10 x origin x year	Supplier share of buyer-product x depreciation	0.005 (0.004)
D bargaining shares	exporter x HS10 x year	Buyer share of supplier-product x depreciation	0.006 (0.007)
D bargaining shares	exporter x HS10 x year	Supplier share of buyer-product x depreciation	0.005 (0.010)
E rel size vs scale	HS10 x origin x year	Log relationship size x depreciation	0.001 (0.001)
E rel size vs scale	HS10 x origin x year	Importer-product scale x depreciation	-0.000 (0.000)
E rel size vs scale	HS10 x origin x year	Exporter-product scale x depreciation	-0.001 (0.000)

Notes: Standard errors in parentheses.

*, **, and *** denote significance at the 10, 5, and 1 percent levels.

Notes. Regressions of annual changes in relationship-level dollar unit values on depreciation interactions, yearly 2009–2020 panel. A depreciation year is one in which the peso depreciates by more than five log points against the dollar. “Top relationship” indicates the top 0.5 percent of relationships by lagged import value within a year. “Long relationship” indicates relationships observed for at least three prior consecutive years. “Exclusive supplier–product” indicates that the foreign supplier sells the HS10 product to a single Colombian buyer in that year.

5 Conclusion

This paper studies imported inflation at the level of supplier–buyer–product relationships. In Colombia, import expenditure is sufficiently concentrated that the price changes of large relationships matter for the aggregate index. The covariance between the simple granular residual and aggregate import-price inflation is equal to 63 percent of the variance of aggregate inflation. I show that this granular component has two sources. About three-quarters reflects the product exposure of large relationships: they are concentrated in products that experience large common price movements. The remainder arises within products, because large relationships reprice differently from other relationships importing the same HS10 good.

The two margins differ across sectors and over time. Product exposure is particularly important for commodities, where import value is concentrated in relationships trading products with volatile common prices. Within-product variation is more important for differentiated goods, with energy as a notable exception. Neither margin necessarily amplifies inflation. During the 2014–16 peso depreciation, both on average reduced peso import-price inflation relative to the price change of the average relationship. In other episodes, they moved in the opposite direction. Concentration therefore determines the weight placed on particular price movements, but not their sign.

These results also clarify what is lost when border prices are measured only at the product level. Product indexes retain the effect of large relationships insofar as those relationships are concentrated in particular products, but they average out price variation across suppliers and buyers within the same product. Preliminary evidence from annual data suggests that this within-product component is associated more with variation on the cost side of the price equation than with changes in model-implied bilateral markups, although that interpretation is necessarily model-dependent. More broadly, the results show that the supplier–buyer relationship is a relevant unit for understanding how external price shocks enter aggregate import prices. Future work can use this margin to study which shocks generate within-product repricing, how sourcing relationships adjust to them, and how those adjustments affect the transmission of global shocks to domestic inflation.

References

- ALVAREZ-BLASER, S., R. AUER, S. M. LEIN, AND A. A. LEVCHENKO (2025): “The Granular Origins of Inflation,” NBER Working Paper 33404, National Bureau of Economic Research.
- ALVIAREZ, V. I., M. FIORETTI, K. KIKKAWA, AND M. MORLACCO (2025): “Two-sided market power in firm-to-firm trade,” *NBER working paper*, #31253.
- AMITI, M., O. ITSKHOKI, AND D. E. WEINSTEIN (2024): “What Drives US Import Price Inflation?” *AEA Papers and Proceedings*, 114, 106–111.
- AUER, R. A., A. A. LEVCHENKO, AND P. SAURÉ (2019): “International Inflation Spillovers through Input Linkages,” *Review of Economics and Statistics*, 101, 507–521.
- BERNANKE, B. S. AND O. J. BLANCHARD (2025): “What caused the US pandemic-era inflation?” *American Economic Journal: Macroeconomics*, 17, 1–35.
- BERNARD, A. B. AND A. MOXNES (2018): “Networks and Trade,” *Annual Review of Economics*, 10, 65–85.
- BURSTEIN, A. T., V. M. CARVALHO, AND B. GRASSI (2025): “Bottom-Up Markup Fluctuations,” *Quarterly Journal of Economics*, 140.
- CONLON, C., N. MILLER, T. OTGON, AND Y. YAO (2026): “Did profits cause inflation?” *International Journal of Industrial Organization*, 105, 103257.
- DI GIOVANNI, J., C. KALEMLI-ÖZCAN, A. SILVA, AND M. A. YILDIRIM (2024): “Pandemic-Era Inflation Drivers and Global Spillovers,” Working paper, National Bureau of Economic Research.
- DI GIOVANNI, J., A. A. LEVCHENKO, AND I. MEJEAN (2014): “Firms, Destinations, and Aggregate Fluctuations,” *Econometrica*, 82, 1303–1340.
- DI GIOVANNI, J., A. A. LEVCHENKO, AND I. MEJEAN (2022): “Foreign Shocks as Granular Fluctuations,” *Working Paper*.
- FELLEGI, I. P. AND A. B. SUNTER (1969): “A Theory for Record Linkage,” *Journal of the American Statistical Association*, 64, 1183–1210.

- FONTAINE, F., J. MARTIN, AND I. MEJEAN (2020): “Price Discrimination Within and Across EMU Markets: Evidence from French Exporters,” *Journal of International Economics*.
- GABAIX, X. (2011): “The Granular Origins of Aggregate Fluctuations,” *Econometrica*, 79, 733–772.
- GLOVER, A., J. MUSTRE-DEL RÍO, AND A. VON ENDE-BECKER (2023): “How much have record corporate profits contributed to recent inflation?” *Federal Reserve Bank of Kansas City Economic Review*, 108.
- KONCZAL, M. AND N. LUSIANI (2022): “Prices, profits, and power: an analysis of 2021 firm-level markups,” *Roosevelt Institute*.
- MARTIN, J., I. MEJEAN, AND M. PARENTI (2023): “Relationship stickiness, international trade, and economic uncertainty,” *Review of Economics and Statistics*, 1–45.
- MONARCH, R. AND T. SCHMIDT-EISENLOHR (2020): “Longevity and the value of trade relationships,” *mimeo, FED Board*.
- PIERCE, J. R. AND P. K. SCHOTT (2009): “A Concordance between Ten-Digit U.S. Harmonized System Codes and Sic/Naics Product Classes and Industries,” *NBER Working Paper*, 15548.
- RAUCH, J. E. (1999): “Networks versus markets in international trade,” *Journal of International Economics*, 48, 7–35.
- WEBER, I. M. AND E. WASNER (2023): “Sellers’ inflation, profits and conflict: Why can large firms hike prices in an emergency?” *Review of Keynesian Economics*, 11, 183–213.

Appendix

A Accounting across frequencies

Table A.1 reports the three-part granular accounting at each calendar frequency and change horizon used in the robustness analysis. It shows that the baseline within-product relationship share is similar when yearly inflation is measured from monthly or quarterly observations, while shorter change horizons contain more within-product variation.

Table A.1: Accounting across calendar frequencies and change horizons

Calendar and change horizon	Average rel.	Granular product	Within product	Simple granular	Periods
Monthly, one-month	15.2	50.2	34.5	84.8	299
Quarterly, one-quarter	26.6	57.8	15.7	73.4	99
Monthly, twelve-month	32.7	50.6	16.7	67.3	288
Quarterly, four-quarter	37.1	46.1	16.7	62.9	96
Annual	40.4	47.5	12.1	59.6	24

Notes. Covariance shares of the average-relationship component, granular product exposure, and within-product relationship variation in equation (7). The last two components sum to the simple granular covariance share; all three sum to 100. Monthly one-month and twelve-month changes use lagged monthly values; quarterly one-quarter and four-quarter changes use lagged quarterly values; annual changes use lagged annual values.

B Panel scale

Table A.2 documents the scale of the monthly customs panel at selected points in the sample and reports the median year. It provides the underlying counts behind the concentration measures in the main text.

Table A.2: Scale of the monthly relationship panel

Year	Rel.-months (m)	Relationships	Suppliers	Buyers	HS10	FOB (\$bn)
2001	0.78	428,666	69,269	16,024	6,025	12.0
2005	1.02	530,212	68,280	17,986	6,213	19.8
2010	1.31	631,584	73,804	21,164	6,313	36.8
2015	1.66	787,054	90,679	26,921	6,446	51.1
2020	1.46	687,859	76,007	25,346	6,493	40.5
2025	2.08	985,823	104,613	29,510	6,615	66.6
Median year	1.61	755,196	84,638	25,936	6,446	43.6

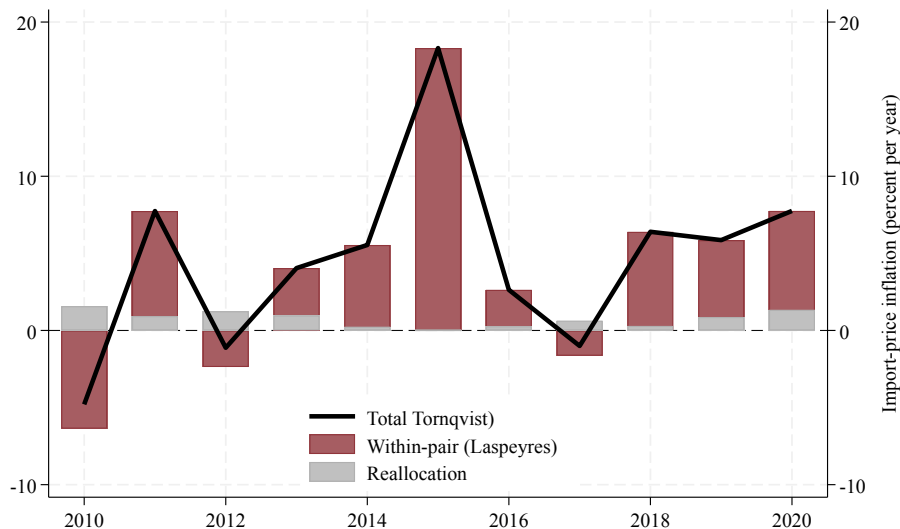
Notes. A relationship is a supplier-buyer-HS10 cell that also holds fixed reported supplier country and origin. Relationship-months count distinct active cells by month. FOB values are in current U.S. dollars. The median row is the median across 2001–2025.

C Additional Figures

The remaining figures provide supplementary diagnostics using yearly observations from the 2010–2020 panel; they are not part of the monthly baseline. Figure A.1 evaluates the construction of the peso import-price index, while the subsequent figures document concentration, volatility, and price-change dispersion in the yearly data.

Figure A.1 asks whether the customs index moves because prices change within continuing relationships or because expenditure shifts across them. The within-relationship Laspeyres component closely tracks the total index, while the reallocation residual is small. Changes in weights among continuing relationships therefore contribute little to the aggregate series.

Figure A.1: Within-relationship Laspeyres component and reallocation residual



Notes. The figure decomposes the relationship-level customs index of peso inflation (thick dark line) into a fixed-weight within-relationship Laspeyres component (maroon bar) and a reallocation residual across continuing relationships (grey bar). The within-relationship component closely tracks the total index, with an annual correlation of 0.99, while reallocation is small and stable, averaging about 1.2 percentage points per year. Relationship entry and exit are excluded by construction.

Figure A.2 compares concentration across relationships, exporters, products, and importers. Panel A.2a reports the import shares held by the upper tail of each distribution, while Panel A.2b reports the corresponding effective numbers over time. The relationship-level basket remains concentrated even relative to other natural units of aggregation.

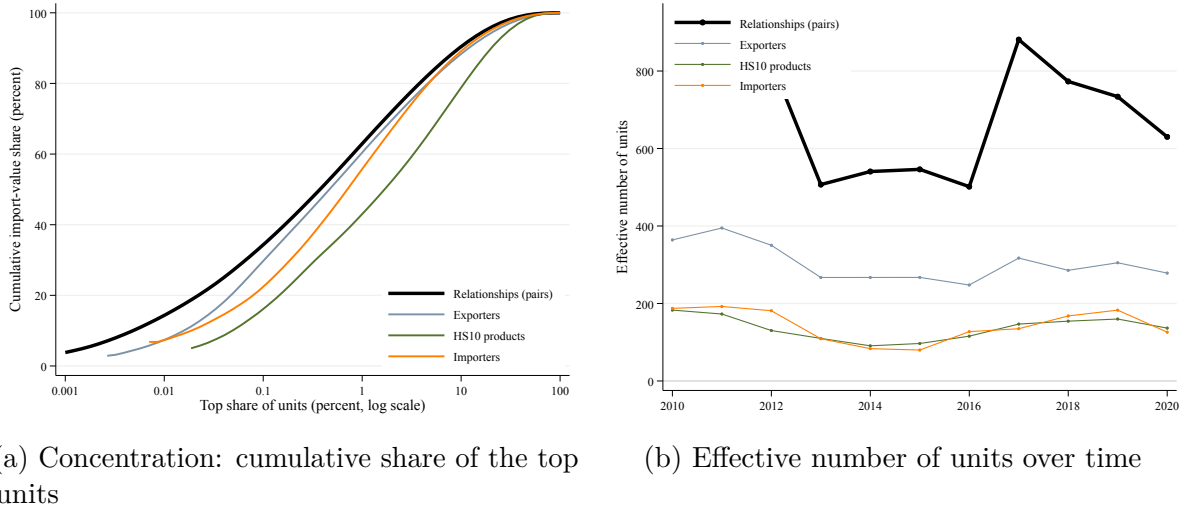
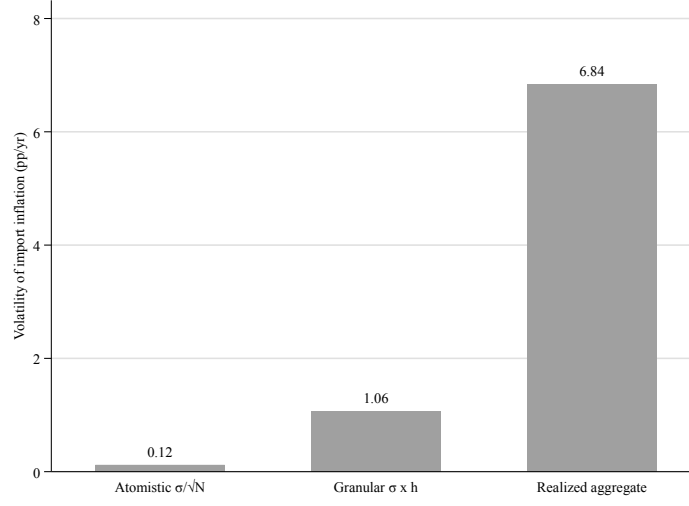


Figure A.2: Concentration and the effective number of import units

Notes. Both panels use lagged-value weights $\omega_{r,t-1}$ and cover continuing relationships, exporters, HS10 products, and importers, 2010–2020. Panel (a): within-year cumulative share of import value held by the top $x\%$ of units (by value), averaged over years. The top 1% of relationships and of exporters each hold about 62–64% of import value, against about 46% for the top 1% of HS10 products. Panel (b): effective number of units $1/\sum_r \omega_{r,t-1}^2$ by year; the import basket prices as though it were an economy of a few hundred equal-sized relationships, against a raw count near 230,000, and this is stable over the sample.

Figure A.3 compares realized aggregate volatility with two benchmarks. The atomistic benchmark assumes equal-sized relationships and therefore predicts extensive diversification. Retaining the observed size distribution and relationship-level shocks raises predicted volatility substantially, but the granular benchmark remains well below realized aggregate volatility. Concentration therefore erodes diversification, although this simple granular term does not by itself account for aggregate volatility.

Figure A.3: Granular versus aggregate volatility of import inflation



Notes. Standard deviation of dollar import-price inflation, 2010–2020. *Atomistic* is $\bar{\sigma}/\sqrt{N}$, the benchmark for a diffuse sector of N equal-sized relationships; *Granular* is $\sqrt{\sum_r \omega_{r,t-1}^2 \varepsilon_{rt}^2}$, the Gabaix term $\bar{\sigma} h$ with heterogeneous relationship shocks; *Realized* is the standard deviation of the aggregate dollar index. Here $\bar{\sigma}$ is the cross-relationship standard deviation of the year-demeaned dollar price change and ε_{rt} the relationship-specific shock.

Figure A.4 documents the underlying dispersion in yearly relationship-level unit-value changes. Panel A.4a shows the full trimmed distribution, and Panel A.4b shows that absolute price changes decline with relationship size. This pattern motivates both the trimming rule and the use of lagged import-value weights: small cells are noisier, but they receive little aggregate weight.

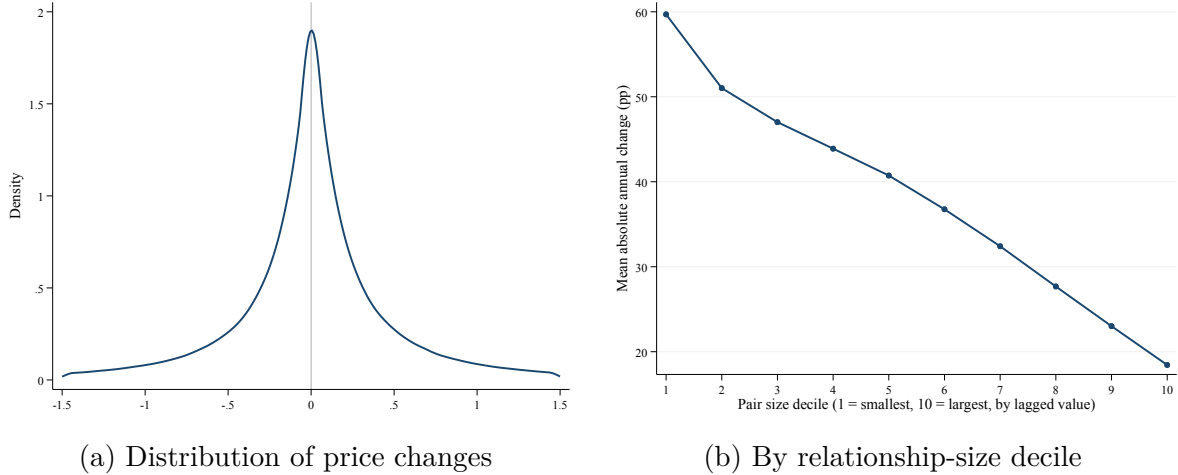
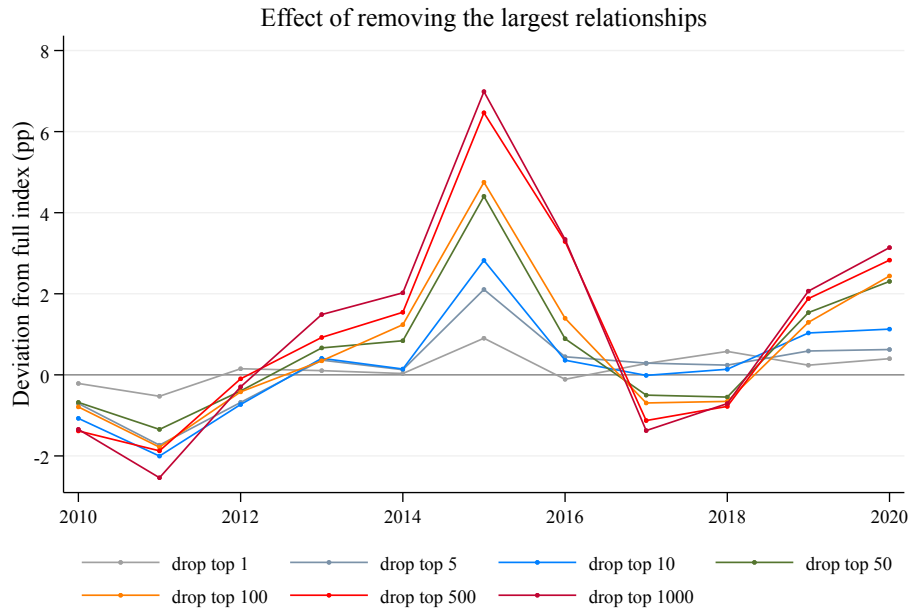


Figure A.4: Dispersion of relationship-level dollar price changes

Notes. Panel (a): kernel density of log year-on-year changes in relationship-level dollar unit values; continuing relationships, changes trimmed to $[-3, 3]$. Panel (b): mean absolute year-on-year change in the dollar unit value, by relationship-size decile (lagged import value).

Figure A.5 provides a direct influence diagnostic by recomputing the import-price index after removing the largest relationships. The index moves when even a small number of top relationships are excluded, confirming that concentration affects the aggregate series rather than merely its cross-sectional description.

Figure A.5: Leave-top- k -out import-price index

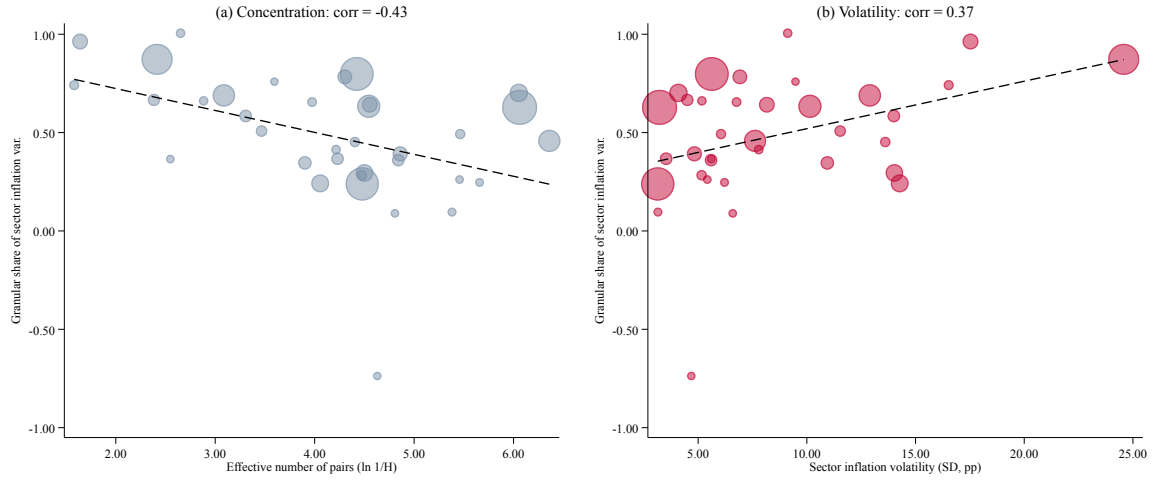


Removing one relationship (of ~230,000) already shifts the index; the top 100 move it markedly, most in the depreciation years. Deviation is currency-i

Notes. Deviation of the dollar (USD) import-price index from its full-sample value when the top k relationships (by lagged import value) are dropped each year, for $k = 1, 5, 10, 50, 100, 500, 1000$. Removing the single largest relationship already shifts the index, and dropping the hundred largest moves it by several percentage points, concentrated in the depreciation years—far more than a diffuse sector of 230,000 units could produce. The deviation is currency-invariant: the common peso/dollar conversion cancels, so it is the same in dollars and pesos.

Figure A.6 examines the same mechanism across HS2 sectors. Granular covariance shares are largest in sectors that combine high concentration with volatile price changes, especially commodity-intensive sectors. The cross-sector evidence supports the distinction in the main text between concentration and the product-price movements to which large relationships are exposed.

Figure A.6: Granularity and volatility across HS2 sectors



Notes. Each point is an HS2 sector; marker area is proportional to sector import value. The granular share is $\text{Cov}(\Gamma^s, \pi) / \text{Var}(\pi)$ within the sector over 2010–2020, with supplier–buyer–product relationships as units. Panel (a) plots the granular share against the log effective number of relationships, $\ln(1/H)$; lower values indicate greater concentration. Panel (b) plots the same share against the sector's standard deviation of dollar import inflation. Granularity is largest in concentrated and volatile commodity sectors, especially oil, fats, and base metals.

D Data Appendix

This appendix records the data-construction details behind the relationship-level panel: the longitudinal exporter identifier that defines continuing relationships, the harmonization of product codes across Harmonized System revisions, panel coverage and continuation, and the exchange-rate series used to convert dollar customs prices into pesos.

D.1 Constructing a Longitudinal Exporter Identifier

Colombian importers are identified throughout by their tax number (NIT), but foreign exporters appear in the customs declarations only as a free-text name, truncated to forty characters, together with a country, city, and address. To build a panel in which the same foreign firm carries the same identifier in every month it appears, I solve an *entity-resolution* (record-linkage) problem on these noisy strings without ground truth, following the probabilistic-linkage framework of Fellegi and Sunter (1969) and its applications to trade micro data (e.g., Bernard and Moxnes, 2018; Monarch and Schmidt-Eisenlohr, 2020). The construction combines deterministic standardization, blocked probabilistic linkage, LLM-assisted classification of ambiguous candidate matches, and validation against hand-coded links.

The pipeline first repairs systematic noise in the name field: corrupted accented characters are mapped back and transliterated to ASCII, text is upper-cased and de-punctuated, and legal-form tokens—the largest source of spurious variation—are canonicalized from an iteratively built dictionary. Deterministic passes within origin country then resolve the bulk of the data: exact standardized names, prefix rules for names severed by the forty-character truncation, token-set rules for word re-orderings, and country repair for records with missing or miscoded codes. The remaining within-country variation is resolved by fuzzy matching: candidate pairs are generated within blocks based on country, name tokens, phonetic keys, and character q -gram nearest neighbors; each pair is scored with Jaro–Winkler, token-set, and TF–IDF character-trigram similarities, augmented with relationship evidence from common importers and products; ambiguous candidates are classified with an LLM-assisted procedure using raw and standardized names together with location and relationship information. Accepted matches are clustered with a union-find structure, and low-cohesion clusters are audited and split.

Acceptance thresholds are chosen on a hand-labeled benchmark of roughly 1,500 candidate pairs sampled stratified by similarity score, targeting precision of at least 0.98; an independent audit of the accepted matches yields a measured precision of 0.992. Identifiers are content-addressed—a hash of the origin country and the cluster’s canonical name—and never reassigned: the similarity model is estimated jointly on a core window of the sample, the resulting raw-tuple-to-identifier crosswalk is frozen and versioned, and tuples outside the core window are assigned by matching against the frozen clusters, with genuinely new firms receiving new identifiers. After harmonization, between 94 and 100 percent of annual import value is assigned to a longitudinal supplier identifier (94–97 percent in 2009–2010, 98–99 percent in 2011–2020, and above 99.7 percent elsewhere), and the adjacent-year linkage rates reported in Table B.3 rise by roughly ten percentage points relative to linking on raw name strings.

D.2 Product Classification across HS Revisions

The 2001–2025 sample spans five revisions of the Harmonized System (HS 1996, 2002, 2007, 2012, 2017, and 2022), and roughly 15 percent of HS6 codes split, merge, or are recoded at each revision. A price change computed across a revision boundary on a recoded product would be lost or contaminated. Following the synthetic-product approach of [Pierce and Schott \(2009\)](#), I link any HS6 codes that ever map into one another in the official UNSD correspondence tables and use the resulting connected components—5,000 families covering 6,569 HS6 codes—as time-consistent product identifiers for across-period comparisons; within-period market definitions use raw HS10 codes. The revision timing is verified in the data: codes unique to each revision first carry import value in exactly 2002, 2007, 2012, 2017, and 2022.

D.3 Panel Coverage and Continuation

The tables in this subsection document the size, coverage, persistence, and cross-sectional structure of the customs panel. Table [B.1](#) reports the annual cross-section of supplier–buyer–product relationships and their aggregate import value.

Table B.1: Firm-to-firm import panel by year

Year	Relationships	Exporters	Importers	HS10	Origins	FOB (\$bn)
2009	520,419	77,858	27,594	7,528	189	39.4
2010	513,546	75,867	27,502	6,695	185	42.1
2011	655,041	86,866	32,535	6,589	178	60.4
2012	689,766	91,961	35,232	6,780	182	58.7
2013	655,580	89,417	32,997	6,594	171	53.2
2014	601,609	86,772	32,181	6,565	184	46.8
2015	663,391	90,945	32,393	6,572	170	52.0
2016	503,153	77,397	28,910	6,412	160	28.8
2017	648,263	87,905	31,663	6,644	169	43.2
2018	485,428	71,969	26,964	6,418	165	29.0
2019	699,676	92,204	32,414	6,662	167	53.7
2020	445,771	68,655	25,856	6,386	154	27.2

Notes. “Relationships” counts distinct exporter–importer–HS10 relationships active in the year (aggregated from the monthly panel); the other columns count distinct exporters, importers, HS10 products, and origin countries. FOB is total customs free-on-board import value (current US\$).

Table [B.2](#) compares the panel’s aggregate import value with official Colombian trade statistics. Because both are built from the same administrative records, the comparison is a reconciliation rather than a coverage check: annual FOB totals match the official DANE series essentially one for one, and to within 0.01 percent in the years where official bulletins permit an exact comparison.

Table B.2: Panel import value versus official merchandise imports

Year	Panel (\$bn)	Official (\$bn)	Ratio (%)
2009	39.4	32.9	119.8
2010	42.1	40.5	104.0
2011	60.4	54.2	111.4
2012	58.7	59.0	99.4
2013	53.2	59.4	89.6
2014	46.8	64.0	73.0
2015	52.0	54.1	96.2
2016	28.8	44.9	64.1
2017	43.2	46.1	93.8
2018	29.0	51.2	56.6
2019	53.7	52.7	101.8
2020	27.2	43.5	62.6
Total	534.4	602.5	88.7

Notes. “Panel” is total FOB import value in the relationship-level panel; “Official” is the DANE merchandise-imports FOB series.

Table B.3 uses the exporter identifier to show how much of each year’s trade can be followed from the previous year, at the exporter level and at the full supplier–buyer–product level.

Table B.4 reports the distribution of relationship-level shipments and unit values used in the price-index construction.

Table B.3: Exporter linkage and relationship continuation, year on year

Year	Exporter present $t-1$		Relationship traded $t-1$	
	% value	% count	% value	% count
2010	91.2	81.6	70.8	36.9
2011	89.9	80.1	70.7	32.2
2012	91.9	82.2	69.2	35.6
2013	91.4	82.4	75.2	38.2
2014	91.6	83.8	73.9	40.4
2015	91.4	83.2	75.8	37.9
2016	92.9	84.1	78.5	44.0
2017	93.3	82.5	75.6	35.1
2018	91.4	85.1	77.3	44.9
2019	91.8	81.4	78.1	34.5
2020	94.1	88.6	82.5	52.7
Average	91.9	83.2	75.2	39.3

Notes. “Exporter present $t-1$ ” is the share of year- t import value (resp. relationships) whose exporter, under the harmonized identifier, also appears in $t-1$. “Relationship traded $t-1$ ” is the share of year- t value (resp. relationships) in exporter–importer–HS10 relationships that also traded in $t-1$.

D.4 Exchange-Rate Series

Exchange rates are monthly averages of the official peso–dollar market rate (*tasa representativa del mercado*, TRM). Each origin currency is converted to pesos through the dollar cross-rate, $e_m^{cop/o} = e_m^{cop/usd} \times e_m^{usd/o}$, which separates the common peso/dollar rate from origin-currency/dollar rates. The sample spans several large, peso-wide swings (Figure B.1). The peso depreciated through the early 2000s to about 2,950 per dollar in early 2003, then appreciated through the commodity boom to a trough near 1,770 in mid-2012. The collapse in oil prices from late 2014—oil is roughly half of Colombian exports—reversed this sharply: the peso lost about 37 percent against the dollar between 2014 and 2015 and roughly 70 percent from the 2012 trough through early 2016. After stabilizing near 2,950 in 2017–2018, it weakened through the COVID-19 shock (near 3,990 in April 2020) and again in the global tightening of 2022, when the monthly average peaked near 4,930 in late 2022 before easing below 3,800 by the end of 2025. Because these are movements of the peso against all currencies at once, the partner series in Figure B.1 co-move, and the common peso/dollar rate accounts for the bulk of measured peso import inflation, while origin-currency/dollar terms contribute little.

Table B.4: Cross-sectional summary statistics

	Mean	SD	p10	p50	p90
Unit value (USD/kg)	446.42	59221.52	1.33	15.84	213.50
Transaction value (000 USD)	75.47	2616.57	0.06	2.05	58.68
Suppliers per buyer-product	3.20	5.34	1.00	2.00	7.00
Buyers per supplier-product	2.44	15.32	1.00	1.00	4.00
Within supplier-product SD of log unit value	0.49	0.54	0.05	0.33	1.14

Notes. Distribution across relationship-month observations, 2001–2025. Unit values and transaction values are right-skewed, so means exceed medians. Partner counts are computed within HS10 product-periods. The last row reports, for each supplier-product cell served by at least two buyers, the standard deviation of the log unit value across those buyers; single-buyer cells provide no within-cell comparison.

Figure B.1: Exchange-rate changes against Colombia's main import partners



Notes. Each line is the log change in the peso price of an origin currency (COP per unit of origin currency) for an origin supplying more than 3.6% of import value (Panama, which invoices in dollars, is excluded); the bold black line is the import-weighted average. The series co-move because they share the common peso/dollar swings that dominate the period.